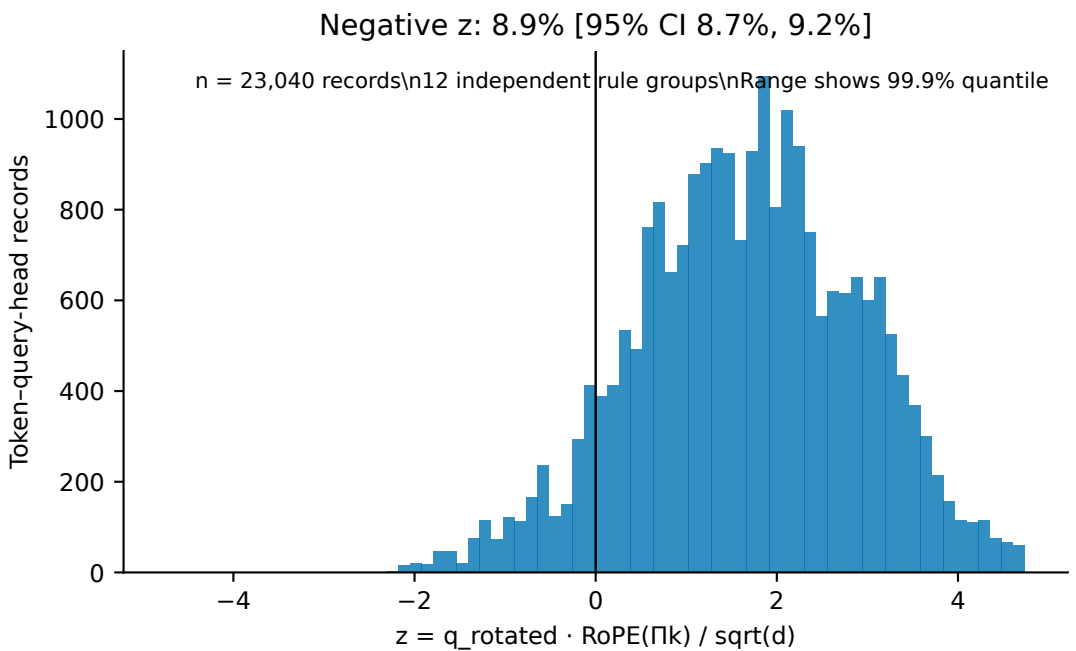
Negative z: 8.9% [95% CI 8.7%, 9.2%]
n = 23,040 records\n12 independent rule groups\nRange shows 99.9% quantile
Token–query-head records
0
200
400
600
800
1000
−4
−2
0
2
4
z = q_rotated · RoPE(Πk) / sqrt(d)